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Research on bilingual language processing and how languages are represented in the mind

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ABSTRACT

Bilingual ability arises naturally in young children exposed to two languages, while second language learning among children and adults often results in bilingual ability as well. *The study of bilingual language processing*, by Nan Jiang (2023), goes beyond what the title announces to take up the related questions of how knowledge of two languages is represented. The study explores the nature of the underlying linguistic structures and how they interface with other cognitive domains. Thus, research on both aspects of bilingualism helps us better understand language development in general. This review essay will examine some of the broader implications for psychological science that the author suggests. The findings from the research are extensive, and the following discussion will propose the possibility of a partial convergence on questions that have been the subject of debate among specialists.

1. Introduction

Research problems in cognitive development greatly benefit from studying interactivity, the kinds of interface that link components, and by focusing on different aspects of the interaction in each case. Professor Nan Jiang's (2023) book chose the relationship between language and (general) cognition, to then narrow the discussion to the connections between first language (L1) and second language (L2) in bilingualism. The promise was that the examples of processing within networks and at their intersections would be especially informative because specific examples help reveal more general relationships and tendencies. The aspects chosen were representation and processing, only the second one, in the interest of conciseness, mentioned in the title. The distinction between the two, representation and processing, nevertheless, is important to keep in mind throughout the book.

2. Summary of the chapters

2.1. Chapter 1: introducing bilingual processing research

The introduction serves as a survey of the subfield of bilingual studies that takes the methods of cognitive science as its starting point. One of these approaches to the question of how language and thought are engaged is to consider the actual physical architecture of each internally diverse system and the corresponding linkages between them. The proposal regarding the basic design features of the entire confederation of underlying structures, "the human mind," is to study it as a "capacity-limited processor" (p. 6). This of course is only one perspective, among others, for the study of the mind. The information to be processed, in this case, comes in two formats,

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following the assumption that they are not the same. The two formats correspond to: linguistic knowledge of two or more languages, and meaning.

Representation, in chapter 1, refers to the idea that the developmental outcomes of acquisition and learning give form to a material implementation in the mind of the different competencies for performing observable abilities. Representations are actual knowledge structures. This definition is carried forward through to all the subsequent chapters. Thus, researchers consider the emergence of linguistic knowledge structures that correspond to:

- mother tongue,
- native-like competence,
- L2 learner-language, and
- asymmetric competence, the result for example of a transition of dominant-language from a first language to a second language, as when an L2 becomes the dominant-language "at the expense" of the original L1. The new dominant language system "displaces" the former L1 system, in part or completely.

The presentation of the research literature in this book takes these categories as cognitively grounded. Along another dimension, the chapter presents the first of a series of differentiations and hierarchies, for example, between: the lexical level, linked to the grammatical knowledge of L1 and L2—*language*, and conceptual structure, or structures—*meaning*.

Another example of an important differentiation is from within the language system itself: the knowledge of phonology is arguably of a different kind than that of the other language knowledge sub-components. According to this line of argument, phonological competence is a specialized component of linguistic competence, separate, for example, from syntax/morphology. According to the hypothesis, its underlying cognitive structure would be largely or entirely domain-specific to one aspect of language (Chabot, forthcoming). Related to this proposal is the evidence of dissociation, between phonology and other kinds of language knowledge, developmentally (Friedmann & Rusou, 2015).

The laboratory-based research on *processing*, familiar to psychometricians working in the other domains of cognition, will directly inform the questions about mental architecture, questions of *representation*. The most interesting will be the controlled experiments, involving bilingual speakers, of language switching. The results then can be compared with those from the grammatical analysis of codeswitching, largely based on the analysis of naturalistic speech samples. On this point, if readers feel uncomfortable with the term "switch," or "switching," for its connotation of "on-off," an alternative would be "shift." In this way, controlled experiments and spontaneous samples of unprompted performance should complement each other. Are the data from each method consistent and representative, in particular considering the data from naturalistic speech samples? To what extent do they serve as evidence to confirm or disconfirm the respective interpretations? These questions don't come up, as much, in the author's discussion of the controlled experiments, for good reason. They will come up more often when considering the comparison with data from qualitative research, a comparison that in the end should be productive.

The first chapter ends with a summary of the early research on the cognitive underpinnings of bilingualism, L2 processing and assessment paradigms, bringing us to the 1990s. The debate revolved around the observation of interaction between L1 and L2, dividing researchers who favored an "independent (or separate)" organization or an "interdependent (or integrated)" organization of the "bilingual's two lexical systems" (p.18). As the following chapters will show, the lines of this debate, while at first appearing to be clear, turn out not to be so.

2.2. Chapter 2: lexico-semantic organization in bilinguals

This chapter begins the roller coaster exposition of studies from the massively productive lab-based work spanning many decades, summarily outlined in the previous chapter. The survey starts with the suggestion of a "[consensus] among many researchers" (p. 44), broached in the first line of the chapter (p. 25). But the expectation of partisan readers, hoping for a smooth ride that confirms their favored model, will be dashed. Many, as this reader does, will try to detect the author's leaning, but will be given no mercy in the remaining pages. A comprehensive assessment of the state of the field will present the full range of what could be said are, or appear to be, the mixed findings. Considering the interface between language ("word") and meaning ("concept"), between the domains of linguistic competence and conceptual structure:

(1) are there, in fact, for bilinguals, two separate lexicons, together with their respective L1 and L2 grammars, in turn connected to a single, shared, conceptual structure? -or-
(2) are concepts language-specific, tied to their corresponding language, to the language that they "are linked to" or "belong to"? -or-
(3) is there one single unitary-integrated lexicon, in this case in correspondence to an integrated and unitary conceptual structure?
(4) Should research dispense altogether with the notion of a mutual autonomy between what are being called the domains of language and thought, the network of the linguistic subcomponents and the knowledge web of concepts? This last question is not explicitly posed by the author. But many of the studies cited do suggest that it deserves consideration, as the problem of language-thought autonomy or integration is as complicated as it is unsettled, at least in some of the details.

The review of the evidence, featuring detailed accounts of how the experiments weigh in on each question, through the remaining chapters to the very end, follows a method: to lay bare the weakness or incompleteness of every side in the debates. No one hypothesis will be vindicated, and a careful reading will allow readers to appreciate how the chapters describe each finding of research objectively. To say that the reading must be deliberate and methodical beginning with the third section of Chapter 2 is an understatement. None of the sections should be omitted. Keeping both eyes on the seemingly conflicting data slows us down. In the end, we are happy that it does because this way of presenting the findings allows us to reflect more carefully on the opposing hypotheses.

Before starting Chapter 3, readers should mark Figure 2.2(d) on page 46, redrawn here as Fig. 1. Aspects of the main idea, more or less, depicted in the Revised Hierarchical Model (RHM) (Kroll & Stewart, 1994) are at the center of the discussion and controversy from this point on. To recap, take note of the proposals (1)—(4) above about the relationship between language ("word") and meaning ("concept").

2.3. Chapter 3: cross-language priming

We begin here with the phenomenon of asymmetry, revealed by different measures of performance, between L1 and L2. As Fig. 1 suggests, asymmetry, or imbalance, is a common outcome of bilingual development. Alternatively, the two language subsystems of a bilingual speaker could be equivalent, both complete and fully formed. The "default condition" in the cross-language priming studies is that the L1 speaker has maintained native-like competence; see the above mentioned "developmental outcomes." The contrasting cross-language priming effect, by many accounts, is consistent with the RHM as the effect is generally more robust in the direction from the stronger L1 to the weaker L2. Links to conceptual structure are, overall, by all accounts, more complete or more secure from L1, consistent with the results.

There appears in the concluding sections of this chapter an interesting, albeit tentative, take-away proposal. In the discussion on the difference between L1 and L2 regarding the strength of the connection between words and concepts, the RHM assumes a "shared conceptual system" (p. 99). The idea of "shared" follows from the hypothesis, hinted at above, that the mental representations that underlie *linguistic knowledge* are of a different kind than those that underlie *concept knowledge*. This proposal, of course, is controversial, making for more interesting reading in the remaining chapters.

2.4. Chapters 4 & 5: selectivity in lexical access & language switch and control

This part of the book provides a contrasting point of view. If the finding of asymmetry in priming and translating could be taken as generally in keeping with the two-lexicons-and-shared-conceptual-knowledge schema (1), the next two chapters now appear to tip the balance in favor of one of the three alternatives (2), (3) or (4). A large number of projects over the many years have examined the different kinds of switching between L1 and L2. In turn, the selectivity question in bilingual production and comprehension performance has been studied just as extensively. The question that can be asked here is: How "permeable" is performance in L1 or L2 to influence or interference from the other language? The suggestion seemed to be that if the result was *not permeable*, i.e., selective, the interpretation would support "separate" mental representations and processing. In contrast, significantly *permeable* and non-selective processing would argue against autonomy of the language domains in relation to each other. If the non-target language (e.g. L2) is always "on," available for activation when speaking or listening in L1, might this evidence suggest a single unitary lexicon, and an integrated and internally homogenous grammar: that is, no autonomous systems or subsystems?

The concluding section of Chapter 5 (pp. 186—187) offers a possible way out of the problem. It proposes to accept the view that the two language subsystems are, in one way or another, always activated. Then the task that bilinguals face, in production and comprehension, is that of control and regulation of activation, managing the competition in selecting one subsystem or the other. Inhibitory processes intervene systematically to regulate selection, for example at a high setting for some kinds of situation (speaking to a group of monolingual listeners), at a low setting for other kinds.

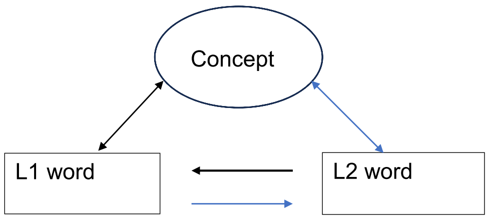

Fig. 1. The Revised Hierarchical Model (Kroll & Stewart, 1994). Note, first of all, that interface relationships maintain interaction between conceptual structure (Concept) and the bilingual lexicon (both L1 word and L2 word). Furthermore, the RHM marks an asymmetry between L1 and L2. While self-evident on one level, in the case of beginner-level and intermediate L2 learning, a number of interesting predictions follow. According to RHM, a stronger connection obtains between concepts and the fully formed first language system (the double-headed black arrow). This would be the case involving language abilities, such as translation. In this regard, note the black and blue arrows in the interaction between L1 and L2. The double-headed black arrow suggests a stronger interface and the blue a weaker interface. The single-headed black arrow suggests a stronger connection (in that direction), the blue a weaker connection.

2.5. Chapter 6: bilingualism beyond lexical processing

The conclusion features the interesting studies on autobiographical memory, how narratives, for example, are more vivid and emotionally intense "when the encoding language and retrieval language are the same" (p. 190). While much of the empirical results along these lines are both predictable and potentially explained by opposing theoretical models, the detailed description is valuable. Some of the sweeping conclusions do however call our attention, for example, from prominent researchers on bilingualism: that memory and experience are stored in or linked directly to the language. In parallel to the discussion on lexical knowledge, the chapter considers the question of syntactic knowledge, "separate" representation of the two language sub-systems or "overlapping." As in the other domains, it's important to keep in mind that "separate" does not imply non-interacting, as Cross Linguistic Interaction applies to both models, concluding observation of this section (6.3). The interesting concluding section on language development and general cognitive consequences of bilingualism ends the book on a broader question that came up earlier, that of the relationship between language competence and conceptual knowledge.

3. First and second language as subsystems

The summary of the chapters now provides a good opportunity at this point to look back on the debates and propose a way forward that in fact the book has hinted at from the beginning. In addition, readers will have noticed by now that it's possible to zoom out from the details of L1-L2 asymmetry, selectivity, priming effect, etc. Some of the "big questions" of cognitive science and the philosophy of knowledge related to the relationship between language and thought hover over the assessment of results. They hover, in particular, over Figure 22(d) of RHM. How should it be better understood, or perhaps just set aside as outdated?

In regard to the rich discussion on selective/non-selective access and switching between the L1 and L2, there appears one model that is a prime candidate for discarding: the one that would favor the view of a strong encapsulation of each language subsystem. The hypothetical proposal of encapsulated representations or components implies that access to one language subsystem (L1 or L2) always closes off access to the other during comprehension and production. We can now discard this view of bilingual processing because access can be either selective or non-selective, depending on the circumstances.

At the same time, evidence for non-selective processing would not bear on the question of overall mental architecture, as it is actually compatible with both so-called separate and integrated-unitary representation hypotheses (p. 106). This is because, despite the connotation of "separate," the subsystems of the larger language system do not require encapsulation: only one way in and one way out. Evidence for non-selective processing *does* contradict, as should be apparent, hypotheses that confine the L1 and L2 subsystems within non-interacting compartments, or components between which interaction is minimal or highly restricted. For now, the "non-interacting compartments hypothesis" has been discarded unless new evidence comes forward.

In the interest of reconciling what appear as inconsistent results, we want to keep our lens zoomed-out to get the bigger picture. A proposal that conceives of highly self-contained L1 and L2 domains, reducing Cross Linguistic Interaction (CLI) to a minimal expression, could perhaps be taken as a "separate systems" hypothesis of the above-mentioned compartments type. But, assuming that L1 and L2 form by necessity an integral part of a single language faculty system, the findings of "non-selectivity," interaction, and mutual influence on performance measures become important to consider. Interaction and mutual influence can occur between subsystems that are autonomous because they interface with other subsystems. How do the different instances of CLI come on-line according to the different intervening factors, aspects of context, and so forth?

4. Another kind of cross linguistic interaction – bilingual speech

In the discussion of Chapter 1, the research on codeswitching came up as a source of evidence for better understanding the concepts related to interface and interactivity. Consider the following hypothetical example below of how borrowing and language switching within sentences tend to be constrained by the grammars of both L1 and L2.

For example, a French-English bilingual speaker has the choice between two kinds of language contact grammatical pattern when commenting (hypothetically) on the book under discussion in this review.

(a) This book is a **veritable tour de force** from the field of bilingual studies.
(b) This book is a ***tour de force véritable*** from the field of bilingual studies.

In (a) an historic borrowing from French, "veritable," is fully integrated into the English lexicon, with the same meaning in both languages. Similarly, the phrase "tour de force" has been borrowed intact with the meaning of "feat demonstrating mastery." Thus, sentence (a) is a sentence in English. In (b), however, the sentence would be a true example of bilingual speech.

The choices for our hypothetical bilingual speaker are:

- treat both "veritable" and "tour de force" as borrowed words or phrases integrated into English as in (a).
- effect an actual code-switch to French as in (b) at the point where "*tour de force véritable* " is inserted.

As an English word (borrowed from French and anglicized) in (a), there will be a strong tendency to give "veritable" an English pronunciation, as it appears in an English grammatical frame (adjective-noun), and because the rest of the sentence is in English. In the inserted codeswitched phrase in (b), especially as it would be rendered according to correct French grammar, "véritable" would be given a French pronunciation. In fact it would be very difficult for a speaker of French to do otherwise in natural speech. If the sentence were to be written by a literate bilingual speaker, "véritable" would also probably be spelled correctly, with "é" in (b), and correctly with "e" in (a). The research on bilingual speech has shown, as in the above example, how codeswitching involves an actual switch to processing the grammar of the embedded constituent, as in (b), according to the rules of its language system. Additionally, in fluent codeswitching, the embedded constituent—the French phrase—tends to be inserted grammatically, as a "unit," into the host language—English—grammatical pattern (Poplack, 2015). Importantly, fully integrated borrowed words, as in (a), are processed differently. The overriding research question is about how bilinguals process each language, especially when they come into contact in the same sentence.

5. Conclusion: reconciling some of the seemingly disparate observations

Developmentally, L1 and L2 share the same learning and acquisition mechanisms, even though the relative weight of each one, with development, will come to vary.[1] For example in child L1 acquisition, the spontaneous and automatic processes characterize development, although not exclusively. Second language learning, for reasons still not well understood, on average, relies to a much greater extent on domain-general mechanisms and deliberate/conscious learning. This shifting in the participation of the different acquisition and learning mechanisms often varies significantly, according to circumstances, a complex question that must be deferred.

Thus, at the gross anatomical level "Language"—with a capital "L"—of the bilingual is in fact a kind of integrated system. This is the starting point of one of the authors (Michel Paradis) cited often in the chapters, whose proposal attempts to incorporate the greater part of both "sides" of the selectivity discussion. In turn, it sets aside the outlier positions at both extremes. The "languages" of the bilingual, then, form networks or subsystems within the larger system (Paradis, 2007). In this way, continuing research focused on the interface correspondences of CLI should be able to resolve the apparently mixed findings of mutual influence between the L1 and L2 subsystems. It is important to read the Paradis proposal not as a "compromise" between the "separate" versus "unitary/integrative" camps. Rather, it follows from the necessity to account for all of the bilingual aphasia data, in particular the cases involving recovery: findings from both non-selective and selective impairment.

The analysis of selective impairment and recovery then suggests the possibility of parallel mechanisms in typical, non-trauma, bilingual ability. The commonly observed capability, during comprehension and production while in the "monolingual mode" (Grosjean, 2022), of *selective inhibition* of the non-target language, requires a full account of how this control mechanism is implemented systematically. What, if not a network or subsystem of circuits, is being inhibited? This scenario is then contrasted to language processing in a "bilingual mode."

The key follow-up to the research evidence presented in this volume will be the comparison, or reconciliation, first between the range of findings on selective and non-selective processing. The next comparison then is between:

- the separate-and-interacting linguistic systems/subsystems proposal, and
- the internally homogeneous, unitary/integrated, system proposal.

Finally, taking into account the broader research context, findings from three parallel lines of research will inform the debate: (i) the developmental aspects of bilingualism as well as loss of language ability associated with dementia, (ii) trauma-related loss and recovery, and (iii) the non-trauma, normal, displacement of first language competence by second or stronger language development—replacement language development. Of special interest for readers of *Cognitive Development*, a research program of this kind would prioritize acquisition and learning across the life span and L1 or L2 attrition (understood as replacement) on the one hand, and on the other hand, erosion of Language, upper case "L" erosion, as in progressive or sudden impairment. The cited references below can serve as a beginning framework for further discussion of the broader implications of *The Study of Bilingual Processing*.

Note

1. An alternate hypothesis would propose that the acquisition/learning mechanisms for L1 and L2 do not overlap to any degree. Thus, for the former (L1), development is governed exclusively by a specialized and domain-specific acquisition device with no participation of general learning of any consequence.

CRediT authorship contribution statement

Norbert Francis: Writing – review & editing, Writing – original draft, Conceptualization.

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